Week 4 Project Logbook: Full-Stack Frontend Integration, Advanced Pages & Public Cloud Deployment

I. Frontend Architecture & Multi-Tier Page Design

(To be completed during UI/UX structuring)

- **Frontend Framework & Layout Structure (React.js/Vite, Tailwind CSS, Responsive Design):**
- **End-User Pages (Inference & Interaction):**
  - *Prediction Interface & Input Controls (File upload, live recording, parameter tuning):*
  - *Results Visualization (Spectrograms, confidence scores, output displays):*
- **Developer Pages (Training & Management):**
  - *Model Training Controls (Trigger training runs, select datasets, configure hyperparameters):*
  - *Model Saving & Checkpoint Management (Export weights, track version history):*
- **Supporting & Analytical Pages:**
  - *System Statistics & Analytics (Dataset distribution, performance metrics, logs):*
  - *Documentation & About Page (System architecture overview, user guide):*

II. Full-Stack API Integration & Backend Routing

(To be completed during client-server wiring)

- **API Contract & Endpoint Mapping:**
  - *Inference Route (*/api/predict*):*
  - *Training & Saving Route (*/api/train*,* /api/save-model*):*
  - *Analytics & Status Route (*/api/stats*,* /api/health*):*
- **Asynchronous Communication & Payload Handling (JSON, multipart/form-data for audio/files):**
- **State Management & Error Handling (Loading spinners, toast notifications, fallback UI states):**

III. Deployment & Public Hosting Strategy

(To be completed before final release)

- **Local Environment Setup (Docker Compose or parallel local node/python execution):**
- **Public Cloud Hosting Platforms Selected (e.g., Vercel/Netlify for Frontend, Render/Hugging Face Spaces/AWS for Backend & Model):**
- **Environment Security & Production Configuration (CORS policies,** .env **management, hidden credentials):**

IV. Practical Implementation & Verification Checklist

(Hands-on execution log for Week 4)

- [ ] **Frontend Routing & Navigation Setup:** Configured client-side routing (React Router) separating end-user, developer, and analytical views cleanly.
- [ ] **End-User Prediction Portal Built:** Implemented user-facing input components, file upload handlers, and dynamic output visualizations.
- [ ] **Developer Training & Checkpoint Portal Built:** Developed administrative UI panels allowing trainers to initiate model training runs and save output checkpoints.
- [ ] **Supporting Statistics & Analytics Pages Added:** Integrated charts and tables displaying dataset statistics, processing speeds, and system metrics.
- [ ] **Full-Stack API Connection Verified:** Connected all frontend forms and buttons to the Express.js backend, ensuring successful request/response lifecycles.
- [ ] **Local Environment Validation:** Tested the complete full-stack web/mobile responsive application locally without errors.
- [ ] **Public Cloud Deployment Completed:** Deployed the application to public hosts (e.g., Vercel/Render/HF Spaces) and verified live external accessibility.

V. Implementation Notes & Code Snippets

- **Architectural / Integration Challenges Faced:**
- **Core Frontend Component or API Route Code Snippet:**